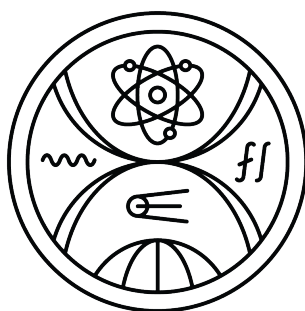
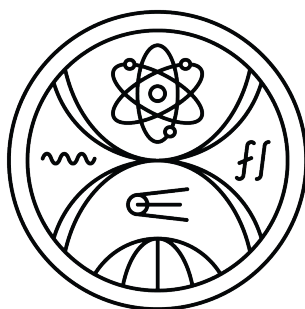


COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



LONGEST INDUCED CYCLES IN GRAPHS WITH BOUNDED DEGREE

Master thesis



LONGEST INDUCED CYCLES IN GRAPHS WITH BOUNDED DEGREE

Master thesis

Study program: Applied informatics
Branch of study: Applied informatics
Department: Department of Applied Informatics
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Študijný odbor: informatika
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Názov: Longest induced cycles in graphs with bounded degree
Najdlhšie indukované kružnice v grafoch ohraničeného stupňa

Anotácia: Štúdium algoritmických a teoretických aspektov dĺžky najdlhšej indukovanej kružnice v grafoch s ohraničeným maximálnym stupňom (najmä regulárnych).

Cieľ: 1. Navrhnuť a implementovať algoritmus, ktorý určí dĺžku najdlhšej indukovanej kružnice v grafe (mal by fungovať aspoň pre 4-regulárne grafy). Odmerať, ako závisí čas behu od veľkosti vstupného grafu.

2. Preskúmať dĺžku najdlhšej indukovanej kružnice v malých 3- a 4-regulárnych grafoch a v špecifickejších triedach grafov (napr. obmedzených obvodom, konektivitou alebo cyklickou konektivitou).

3. Na základe výpočtových výsledkov popísať triedy grafov s extrémnymi dĺžkami indukovaných kružníc (krátkymi aj dlhými).

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.....
študent

.....
vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2027

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Bc. Eliška Krajčiová

Acknowledgement

Thank you

Abstract

Keywords: graph, regular, induced, cycle

Abstrakt

Kľúčové slová: graf, regularny, indukovaný, kruznica

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Terminology

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Chapter 1

Introduction

1.1 Motivation

Induced cycles are a natural object at the intersection of structural and algorithmic graph theory. From the structural point of view, they capture cycle behaviour under a strong restriction: the chosen cycle must be chordless, so the subgraph induced by its vertices is exactly the cycle itself. From the algorithmic point of view, this makes the problem significantly harder than finding an arbitrary long cycle, since even graphs with many long cycles may have only short induced ones.

In this thesis we study the problem of determining the length of a longest induced cycle in graphs of bounded degree, with the main focus on regular graphs. This setting is motivated by several factors. First, regular graphs form a rich and well-structured class in which both extremal behaviour and algorithmic performance can be studied systematically. Second, regular graphs of small degree can be generated exhaustively for moderate orders, which makes them well suited for computational experiments. Third, the author's bachelor thesis focused on the design and implementation of an algorithm for the longest induced cycle problem in cubic graphs, so it is only natural to expand to graphs with bounded-degree, especially regular ones.

This thesis builds directly on the author's bachelor thesis. That work introduced an exact backtracking algorithm based on growing induced paths, maintaining forbidden vertices to avoid chords, and pruning the search space using structural observations and upper bounds. It also included computational experiments on bridgeless cubic graphs and formulated several conjectures suggested by the observed data. [3]

The present thesis extends this line of research in two directions. The first direction is algorithmic: we move from cubic graphs to a broader bounded-degree setting, while still keeping regular graphs as the main experimental domain. The second direction is structural and experimental: we investigate how the length of a longest induced cycle behaves in small regular graphs under additional restrictions such as girth, connectivity,

planarity, or circumference. Particular attention is paid to graphs with extremal behaviour, that is, graphs whose longest induced cycles are unusually short or unusually long.

A secondary motivation of this thesis is to better understand the practical behaviour of exact algorithms on graph instances that are, in principle, similar. Even when all tested graphs have the same order and degree, the running time of an exact algorithm may still vary noticeably depending on finer structural properties of the graph. This suggests that graph parameters beyond order and degree may play an important role in determining algorithmic hardness. For this reason, we also include a short machine-learning-based exploratory study, whose purpose is not to replace exact methods, but to examine whether simple graph descriptors carry useful information about the length of a longest induced cycle or about the empirical difficulty of the computation.

Chapter 2

Related Work

The longest induced cycle problem belongs to a family of graph optimization problems concerned with finding large induced substructures. An induced cycle, also called a chordless cycle, is a cycle such that no two non-consecutive vertices of the cycle are adjacent in the graph. This additional inducedness condition makes the problem substantially different from the classical longest cycle problem, since it is not enough to find many vertices on a cycle; one must also avoid chords between them.

This chapter briefly discusses two works that are particularly relevant to the present thesis. The first one studies exact optimization methods for the longest induced cycle problem itself. The second one investigates induced cycles and induced 2-regular subgraphs from a more structural point of view, with particular relevance to regular graphs.

2.1 Exact Methods for the Longest Induced Cycle Problem

Anaqreh, G.-Tóth and Vinkó studied the longest induced cycle problem from the perspective of exact optimization methods [1]. Their work is directly related to this thesis, since it considers the same underlying optimization problem: finding a longest cycle that is induced by its vertex set. The authors formulate the problem using integer linear programming and compare several exact models for solving it.

The paper proposes three main formulations. The first one is an ordering-based formulation, where additional variables are used to impose an order on the selected vertices and thereby prevent invalid subtour-like solutions. The other two formulations are based on adapting models for the longest induced path problem. In particular, the cycle-elimination model starts from a formulation for induced paths and modifies it so that the selected vertices form one induced cycle instead of a path or a collection of smaller structures. Since some of the required constraints are too numerous to

include explicitly from the beginning, the authors solve parts of their models using a branch-and-cut approach.

This is relevant for the present thesis for two reasons. First, it provides a recent exact approach to the same problem, but in the setting of general graphs rather than regular graphs. Second, it shows that the relationship between induced paths and induced cycles is useful in exact algorithms. A similar idea appears in the algorithmic approach considered in this thesis: instead of immediately constructing a cycle, one can build an induced path and then try to close it into an induced cycle while preserving the induced property.

The difference is that Anaqreh, G.-Tóth and Vinkó express this idea through an integer programming model with binary variables and elimination constraints, while this thesis uses it directly as a graph search principle. Thus, their work can be viewed as a general solver-based exact approach, whereas the present thesis focuses on more specialized algorithms and structural observations for regular graphs.

2.2 Induced Cycles in Regular Graphs

Henning, Joos, Löwenstein and Sasse studied induced cycles in graphs from a structural point of view [2]. Their work does not focus on the longest induced cycle as a single cycle. Instead, they consider the maximum order of an induced 2-regular subgraph of a graph. A 2-regular graph is a disjoint union of cycles, and therefore an induced 2-regular subgraph may consist of several vertex-disjoint induced cycles.

This distinction is important. The parameter studied in [2] measures how many vertices can be covered by a collection of induced cycles, while the present thesis asks for one longest induced cycle. Therefore, their results cannot in general be used directly as bounds for the length of a longest induced cycle. Nevertheless, the paper is relevant because it investigates how large induced cyclic structures must be in graphs with restricted degree, especially in regular graphs.

One of the results in the paper gives lower bounds on the size of induced 2-regular subgraphs in regular graphs. For a k -regular graph with $k \geq 3$, they prove that the graph contains an induced 2-regular subgraph whose order is bounded from below in terms of the number of vertices and the degree. In the cubic case, this implies that every cubic graph contains an induced 2-regular subgraph on more than a quarter of its vertices. They also obtain stronger results for special classes such as cubic claw-free graphs.

For this thesis, the main relevance of this work is that it supports the idea that regular graphs contain large induced cyclic structures. Although the object studied there may be a union of several cycles rather than a single cycle, the results provide

useful structural context. They show that degree restrictions can force the existence of relatively large induced cycle-like subgraphs, which motivates the study of longest induced cycles specifically in regular graphs.

2.3 Position of This Thesis

The two papers discussed above represent two complementary directions. Anaqreh, G.-Tóth and Vinkó approach the longest induced cycle problem through exact mathematical programming formulations for general graphs [1]. Henning, Joos, Löwenstein and Sasse study induced cyclic structures from a structural graph-theoretic perspective, with results that are especially relevant for regular and cubic graphs [2].

The present thesis lies between these two directions. Like the first paper, it is concerned with computing longest induced cycles. However, instead of relying on an integer programming solver, it studies direct algorithmic ideas based on graph search, pruning, and the construction of induced paths that can be closed into cycles. Like the second paper, it focuses on regular graphs and is interested in the structure forced by regularity. However, the objective here is a single longest induced cycle, not a maximum induced union of cycles.

Thus, the contribution of this thesis is to study the longest induced cycle problem in the more specific setting of regular graphs, combining exact algorithmic computation with observations about the structure of the resulting cycles.

Chapter 3

A Machine Learning Perspective on Instance Hardness

This chapter presents a short exploratory machine learning study related to the exact algorithms considered in this thesis. Its purpose is not to replace the exact algorithm by a predictive model. Rather, we use machine learning only as a secondary tool: to test how much information about the problem is already contained in a few simple graph descriptors, and how far such an approach remains from actually solving the problem.

3.1 Motivation

When an exact algorithm is run on many regular graphs of the same order and degree, its running time is often very small in absolute terms, yet still differs by clear constant factors from one graph to another. This suggests that some graph instances are structurally easier or harder for the algorithm, even if they belong to the same broad class. A natural question is whether this notion of instance hardness can be predicted from cheap graph parameters.

There is also a second reason to include such an experiment. If a learning model built only from simple graph invariants were already able to predict the answer very accurately, then this would suggest that the target quantity is largely controlled by a few coarse structural parameters. If, on the other hand, prediction becomes unreliable once graphs get larger or more varied, this supports the view that the longest induced cycle depends on finer structural details of the graph and therefore still requires exact combinatorial methods. In this sense, the machine learning part of the thesis is best understood as an auxiliary diagnostic experiment rather than as a main contribution.

3.2 Initial Feature-Based Experiments

In a previous project, regular graphs generated by `genreg` were used as input instances, and for each graph an exact combinatorial algorithm computed the length of a longest induced cycle together with its running time. The data set contained 43 699 regular graphs, mainly for degrees $k \in \{4, 5, 6\}$ and for small numbers of vertices. For each graph, the recorded features included the regularity, the number of vertices, minimum edge cut, girth, triangle count, number of induced 4-cycles, diameter, and average distance.

Using these features, several prediction tasks were considered: estimating the running time, classifying graphs into easier and harder instances, and estimating the length of a longest induced cycle. The main observation was that simple graph descriptors already capture part of the behaviour. For small graphs, the predictions were often fairly close to the true value, and some features such as the number of triangles, the number of induced 4-cycles, and the average distance appeared to be informative.

However, these findings must be interpreted carefully. On small graphs, the possible values of the longest induced cycle lie in a narrow range, so even an error of one vertex may still look acceptable. This does not mean that the model has understood the combinatorial structure of the problem. It may simply reflect the limited variability of the data.

3.3 Why Larger Graphs Matter

For this reason, the more informative test is to repeat the experiment on larger regular graphs. If the same feature-based approach still works there, this would indicate that a few simple invariants already encode a surprising amount of information. If it works noticeably worse, then this suggests that the descriptors are too coarse and that the exact problem depends on graph structure not visible from those invariants alone.

This also highlights a basic limitation of the approach. The exact algorithm receives the whole graph as input and works directly with its adjacency structure. By contrast, the learning model in our experiment receives only a short list of numerical descriptors. So even in principle, it has much less information available. This is precisely why the machine learning part is interesting: if prediction is close, then those few descriptors already say a lot; if prediction fails, then it reinforces the need for exact graph algorithms.

One could try to enrich the input further, for example by adding the circumference or even by feeding the entire adjacency matrix of the graph to the model. We did not pursue either option here. The circumference is itself computationally difficult to determine exactly, and therefore would already amount to supplying the model with a

strong graph-theoretic parameter closely related to the target. Similarly, using the full adjacency matrix would shift the experiment away from the present question, which is whether a small set of simple graph descriptors is sufficient.

3.4 Experiments on Larger Regular Graphs

To complement the earlier small-instance study, we generated a new data set of larger regular graphs. The instances were grouped by regularity and order, with file names of the form `d_n_tests`, where d denotes the regularity and n the number of vertices. Most classes contained 500 sampled graphs, with the only exception being the class $(6, 20)$, where only 50 graphs were available due to generation constraints.

The resulting data set contained 5600 regular graphs in total. It included cubic graphs on 20, 22, 24, and 26 vertices, 4-regular graphs on 20–24 vertices, 5-regular graphs on 20 and 22 vertices, and a smaller sample of 6-regular graphs on 20 vertices. For each graph we recorded the following graph descriptors:

`{regularity, vertices, min_edge_cut, girth, triangles, c4_induced, diameter, avg_distance}`,

together with the running time of the exact algorithm and the length of a longest induced cycle.

In this larger data set, the average running time increased clearly with regularity: it was approximately 0.27 ms for degree 3, 0.45 ms for degree 4, 1.23 ms for degree 5, and 1.79 ms for degree 6. At the same time, the average length of a longest induced cycle behaved in the opposite direction: for example, it was about 13.56 for $(3, 20)$, 16.81 for $(3, 26)$, 10.57 for $(4, 20)$, 9.62 for $(5, 22)$, and 7.90 for $(6, 20)$. Thus the larger graphs indeed provide a wider and more informative range of target values than the earlier small-instance data.

An additional feature of this data set is that all sampled graphs have girth at least 4. As a consequence, the triangle count is identically zero in this experiment and therefore has no predictive value here. This contrasts with the earlier small-instance experiments, where triangles played a more visible role.

3.5 Prediction Tasks and Models

We considered three simple prediction tasks.

First, we tried to estimate the running time of the exact algorithm from the graph descriptors listed above. Because the running times are very small and vary over several scales, we used their logarithms rather than the raw values.

Second, we considered a coarse distinction between easy and hard graphs, according to whether the running time lies below or above a fixed threshold. In our data this threshold was 0.606 ms, which divided the graphs into 4197 easier instances and 1403 harder ones.

Third, we considered a finer notion of hardness by dividing each fixed (n, d) class into three groups: easy, medium, and hard. This removes the trivial effect that graphs of different size or degree may have systematically different running times.

For these tasks we used only simple standard models. The purpose was not to optimize prediction as much as possible, but rather to see how much can already be inferred from a small number of graph descriptors.

3.6 Results

The first task was the estimation of running time. Here the prediction followed the true behaviour fairly well: broad differences in running time were captured quite successfully, and the model was able to explain a substantial part of the variation between instances. In other words, the descriptors already contain enough information to say which graphs are usually easy and which are usually more demanding.

For the coarse easy-versus-hard classification task, the predictions were very good. The model separated the two classes with high reliability. Among the most informative features were the number of induced 4-cycles, the average distance, the regularity, and the minimum edge cut. This indicates that the algorithmic difficulty is influenced not only by degree and order, but also by both local cycle structure and more global shape of the graph.

For predicting the length of a longest induced cycle itself, the model also performed surprisingly well. On average, the predicted value was usually off by clearly less than one vertex. This means that from a few simple graph descriptors one can already get a rough estimate of the answer.

However, the most important observation comes from the third task, namely the three-way classification into easy, medium, and hard instances within each fixed (n, d) class. Here the prediction quality dropped substantially. The model still did better than random guessing, but the distinction between the three classes became much less reliable. This is exactly the behaviour one should expect if the chosen graph descriptors capture broad trends but not the finer combinatorial structure that determines the exact difficulty of a particular instance.

In this finer setting, the most informative descriptors were the average distance and the number of induced 4-cycles, while the regularity itself played only a very small role. This again supports the idea that once order and degree are fixed, the remaining

hardness is controlled by more delicate structural properties.

3.7 Interpretation

Several conclusions can be drawn from these experiments.

First, machine learning is able to get reasonably close to the target quantity, even when it receives only a short list of graph descriptors instead of the whole graph. This shows that some broad structural features of regular graphs already correlate with the length of a longest induced cycle and with the running time of the exact algorithm.

Second, this does *not* mean that machine learning solves the problem. What it produces is only an estimate, not the exact answer. Moreover, the estimate becomes much less reliable when one asks for finer distinctions between instances of the same order and degree. Thus the learning model captures broad tendencies, but it does not replace a genuine graph algorithm.

Third, there is a notable asymmetry between the two approaches. The exact algorithm receives only the graph itself and from that alone computes the true value. The learning model, by contrast, does not work directly with the graph: it first needs several additional descriptors such as girth, number of induced 4-cycles, diameter, and average distance, all of which must be computed separately. Even then, it still produces only an approximation. From the perspective of this thesis, this is precisely the main lesson of the experiment: machine learning may provide useful heuristic insight, but exact computation still requires a direct combinatorial method.

Finally, the experiment is still informative for the graph-theoretic part of the thesis. It points to graph features that seem to correlate with algorithmic hardness, especially induced 4-cycles and average distance. Even though these observations are not proofs, they may help indicate which graph classes are likely to be easier or harder for exact search.

3.8 Role in This Thesis

The machine learning experiments in this thesis are therefore included mainly for context and interpretation. They show that coarse structural information already allows a rough prediction of both the answer and the running time, but they also show the limits of such an approach. The model gets close, yet it does not truly solve the problem, and it requires many additional parameters while the exact algorithm needs only the graph itself. For this reason, machine learning remains secondary in the present thesis: it offers some insight into instance hardness, but the main emphasis remains on exact algorithms, computational experiments, and structural analysis.

Chapter 4

General Bounds in Regular Graphs

In this chapter we record two general observations that apply to longest induced cycles in regular graphs. We begin with a simple lower bound coming from girth: every shortest cycle in a graph is automatically induced. We then turn to an upper bound for d -regular graphs, which generalizes the cubic bound proved in the bachelor thesis. In particular, for $d = 3$ we recover the bound

$$\iota(G) \leq \frac{3n}{4},$$

already established for cubic graphs in the previous work [3].

4.1 A General Lower Bound from Girth

Recall that the *girth* $g(G)$ of a graph G is the length of a shortest cycle contained in G , and that $\iota(G)$ denotes the length of a longest induced cycle in G . The first observation is immediate but useful as a baseline.

Lemma 4.1. *Let G be a graph containing at least one cycle. Then*

$$\iota(G) \geq g(G).$$

Proof. Let C be a shortest cycle in G . We claim that C is induced.

Indeed, suppose that C had a chord, that is, an edge joining two non-consecutive vertices of C . This chord would split C into two cycles, each strictly shorter than C . That contradicts the choice of C as a shortest cycle in G .

Therefore C is an induced cycle. Since $\iota(G)$ is the maximum length of an induced cycle in G , we obtain

$$\iota(G) \geq |C| = g(G).$$

□

The bound of the previous lemma is usually far from sharp, but it is completely general and already shows that graphs of large girth must contain induced cycles of nontrivial length. In particular, any improvement of the lower bound must necessarily use more information than just the existence of a cycle.

4.2 A General Upper Bound for Regular Graphs

We now derive a general upper bound for the length of an induced cycle in a regular graph. The argument is a direct counting proof and may be viewed as a natural extension of the cubic case.

Theorem 4.2. *Let G be a d -regular graph on n vertices, where $d \geq 2$. Then every induced cycle $C \subseteq G$ satisfies*

$$|C| \leq \frac{d}{2(d-1)} n.$$

Consequently,

$$\iota(G) \leq \frac{d}{2(d-1)} n.$$

Proof. Let C be an induced cycle in G , and write $|C| = k$. Since G is d -regular, every vertex of C has degree d in G . Exactly two of these edges belong to the cycle C , so each vertex of C has precisely $d - 2$ neighbours outside C .

Let

$$N_C = \{ w \in V(G) \setminus V(C) : \exists v \in V(C) \text{ such that } vw \in E(G) \}$$

be the set of vertices outside C adjacent to at least one vertex of C .

Counting the edges between $V(C)$ and N_C from the side of the cycle, we obtain exactly

$$k(d-2)$$

such edges, because each of the k vertices of C contributes exactly $d - 2$ edges leaving C .

On the other hand, every vertex of N_C has degree d in G , and therefore can be adjacent to at most d vertices of C . Hence

$$k(d-2) \leq d|N_C|.$$

It follows that

$$|N_C| \geq \frac{k(d-2)}{d}.$$

Now $V(C)$ and N_C are disjoint subsets of $V(G)$, so

$$k + |N_C| \leq n.$$

Combining this with the previous inequality gives

$$k + \frac{k(d-2)}{d} \leq n.$$

Factoring out k , we get

$$k \left(1 + \frac{d-2}{d} \right) \leq n,$$

that is,

$$k \cdot \frac{2d-2}{d} \leq n.$$

Therefore

$$k \leq \frac{d}{2d-2} n = \frac{d}{2(d-1)} n.$$

Since C was an arbitrary induced cycle in G , this proves that every induced cycle has length at most $\frac{d}{2(d-1)}n$. Taking the maximum over all induced cycles yields

$$\iota(G) \leq \frac{d}{2(d-1)} n.$$

□

Corollary 4.3. *Let G be a cubic graph on n vertices. Then*

$$\iota(G) \leq \frac{3n}{4}.$$

Proof. Substituting $d = 3$ into the general bound gives

$$\iota(G) \leq \frac{3}{2(3-1)} n = \frac{3n}{4}.$$

□

The previous corollary coincides with the cubic upper bound already proved in the bachelor thesis [3]. Thus the theorem above can be seen as its regular-graph generalization.

4.3 Discussion

Together, the two bounds obtained in this chapter give the general estimate

$$g(G) \leq \iota(G) \leq \frac{d}{2(d-1)} n$$

for every d -regular graph G containing a cycle.

The lower bound is completely universal, but usually weak. The upper bound is also general, and becomes more restrictive for small values of d . For cubic graphs, it yields the familiar factor $3/4$. For 4-regular graphs, it becomes

$$\iota(G) \leq \frac{2n}{3},$$

and for 5-regular graphs,

$$\iota(G) \leq \frac{5n}{8}.$$

These inequalities provide a natural starting point for the more refined analysis in later chapters. In particular, the main challenge is not to obtain a universal lower bound—the girth already gives one—but rather to find stronger lower bounds for special graph classes, for example under assumptions on connectivity, cyclic connectivity, planarity, or other structural restrictions.

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